Adrian-Costin MANEA

General Information

Date and place of birth:	3rd of March, 1989, Buzău, Romania
Residence:	Dristor District, Sector 3, Bucharest, Romania
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I am interested in using algebraic methods for the study of noncommutative geometry, which is at the border between modern (quantum) algebra and particle physics, as well as high energy physics.

Holding a Bachelor's degree in Physics, I am familiar with the specific language and tools of study in this domain and, thanks to the Master's degree, followed by the PhD. I hold in Mathematics (Algebra), I studied in great depth mathematical methods and concepts which are fundamental for research in noncommutative algebra and connected domains.

Moreover, other fields that interest me are *science education*, as well as history and philosophy of science in general, and of mathematics, in particular. I work and study to develop software tools for these purposes.

Professional Experience

2017–present

Teaching Assistant
Faculty of Applied Sciences, Department of Mathematical Methods and Models, "Politehnica" University of Bucharest

2016–2017

Research Assistant
Institute of Mathematics of the Romanian Academy
Member in the RU-TE-2014-4-0370 Project, *Harmonic Analysis on Solvable Lie Groups*.
Director: SR2 Dr. Ingrid Beltiță

2013–2016

Teaching Assistant (per hour payment)
Faculty of Mathematics and Computer Science, University of Bucharest
Seminaries of algebra for the first and second undergraduate years, for students minoring in Pure Mathematics, Applied Mathematics, Information Technology and Computer Science.
Contents: Group theory, Ring theory, Arithmetics in integral rings, Modules and vector spaces.

2011–2016

Research Assistant
Research Center in Algebra and Number Theory, University of Bucharest
Member in the Research Grant IDEI 0635, *Hopf Algebras in Category Theory, Representation Theory and Combinatorics*, Director Prof. Dr. Sorin Dăscălescu.

EDUCATION

2013–2016

PhD. in Mathematics
Doctoral School of Mathematics, Faculty of Mathematics and Computer Science, University of Bucharest
Thesis: *Homological Methods in the Study of Algebras and Coalgebras*
Advisor: Prof. Dr. Dragoș ȘTEFAN.
I obtained a POSDRU Scholarship (project 159/1.5/S/137750) for the period June 2014–December 2015, as well as the scholarship from the Doctoral School for the rest of the study period.

2011–2013

Master of Science in Algebra
Faculty of Mathematics and Computer Science, University of Bucharest
Thesis: *Homological Methods in the Study of Noncommutative Algebras*
Advisor: Prof. Dr. Dragoș ȘTEFAN.
Admitted with the general average of 10/10 (written and oral examination), graduated with 10/10 awarded for the dissertation thesis.

2008–2011

Bachelor of Science in Physics
Faculty of Physics of the University of Bucharest, Computer Methods in Physics (taught in English)
Thesis: *Geometric Algebra and the Dirac Equation*
Advisor: Lect. Dr. Roxana ZUS.
Admitted with the general average 9.97/10, awarded 10/10 for the thesis.

2004–2008

Baccalaureate
National College B. P. Hasdeu, Buzău, mathematics and computer science
Baccalaureate average: 9.90/10, July 2008.

RESEARCH

2017

On Koszulity of Finite Graded Posets
J. Algebra Appl. 16, 1750139 (2017), 20 pages
With Dragoș ȘTEFAN

2016

The Ext Ring of a Koszul Ring
Bull. Math. Soc. Sci. Math. Roumanie, Tome 59 (107) No. 1, 2016, pp. 51–63

2016	**Further Properties and Applications of Koszul Pairs** *SIGMA 12 (2016), 092, 24 pages* With Dragoș ȘTEFAN
IN PROGRESS	**The Hochschild Cohomology Ring of a Twisted Koszul Pair**
IN PROGRESS	**Deformation Theory and Koszul Algebras (survey)**
IN PROGRESS	**A Course of Homological Algebra** *With Dragoș ȘTEFAN*

CONFERENCES AND SCIENTIFIC EVENTS

OCTOBER 2017	International Workshop on Graphs, Networks and Digital Humanities *Research Institute of the University of Bucharest* Talk: *The Modern Ubiquity of Networks: From Cells to Ecosystems and Companies*
MAY 2016	Brauer Groups, Hopf Algebras and Monoidal Categories *Università degli Studi di Torino, Italy* Poster: *On Koszulity of Finite Graded Posets*
JULY 2015	Scientific Session for PhD. Students *Faculty of Mathematics and Computer Science, University of Bucharest* Presentation: *Koszul Algebras and Coalgebras*
JUNE 2015	The Eighth Congress of Romanian Mathematicians *"Alexandru Ioan Cuza" University, Iași, Romania*
MAY 2015	Research Stage *LMIA, Université de Haute Alsace, Mulhouse, France* Invited by Prof. Dr. Abdenacer MAKHLOUF
SEPTEMBER 2013	Memorial Conference for J. L. Loday *IRMA, Strasbourg, France*
AUGUST 2013	Anniversary Conference *Faculty of Sciences – 150 Years*
JUNE 2013	The International Meeting of SSMR and AMS *"1 Decembrie 1918" University, Alba Iulia, Romania*
MAY 2013	*Rings, Categories and Hopf Algebras* *Faculty of Mathematics and Computer Science, University of Bucharest* Anniversary Workshop dedicated to Prof. Dr. Constantin NĂSTĂSESCU
OCTOBER 2012	Anniversary Conference for Prof. Dr. Fred van OYSTAEYEN *Antwerp University, Belgium*

Other Courses and Conferences

IMAR

Sorin Dăscălescu (FMI) – *Hopf Algebras, Quantum Groups and Integrals*
Jean Ludwig (Lorraine) – *The C^*-algebra of the Heisenberg Group*

FMI

Marian Aprodu (FMI) – *Homological Methods in Algebraic Geometry*
Vasile Brînzănescu (IMAR) – *Vector Bundles*
Gigel Militaru (FMI) – *Homological and Categorial Methods in the Theory of Hopf Algebras*
Sergiu Moroianu (IMAR) – *Introduction in the Theory of Chern-Simons invariants*
Camil Muscalu (Cornell) – *Calderón Operators*
Dragoș Ștefan (FMI) – *Deformations of Algebraic Structures*

Foreign Languages

English:	Full working proficiency
French:	Average working proficiency
Spanish:	Beginner.

Varia

Pedagogy

I have graduated the Pedagogy Module, BSc. level

Operating Systems

macOS, GNU/Linux, Windows
Experienced user

Programming

C11:	medium level
Python 2 and 3:	medium level
Common Lisp (SBCL), elisp:	medium level
Perl 5:	beginner
HTML, CSS, JS:	beginner
LaTeX:	advanced.

Other Software

I have medium to advanced level of understanding and operation with the following software and tools:

Text editors:	vim, emacs, nano, Sublime Text 3, Atom
Source control:	git and GitHub
System administration:	*nix shells (bash & zsh) use and basic scripting

Philosophy

The Technically Manifolded (Classical and Quantum) Space Ontology
Society and Politics, vol. 11, No. 1, 2017, pp. 91–93
The article reviews the book *The Deep Metaphysics of Space*, Springer 2016, by Ed. Slowik, which treats the problem of (physical and geometric) space, from Newton and Leibniz, to Einstein.

Conceptions of Space in the Scientific Revolution, 1st semester 2015–2016
Dana Jalobeanu, Faculty of Philosophy, University of Bucharest
I participated to the entire course throughout the semester.
My talk: *Elements of Differential Topology in the Leibniz-Clarke Correspondence*

Ontology of Science, 2nd semester 2015–2016
Sorin Costreie, Faculty of Philosophy, University of Bucharest
I participated to the entire course, throughout the semester, as a teaching assistant.
My talks: set theory, physical and mathematical modelling (quantum formalism, structuralism in Algebra and Geometry).

OTHER PUBLICATIONS

Linked, The New Science of Networks, Albert László Barabási
Romanian translation, Brumar, Timişoara 2017, pp. 316
I was a Scientific Consultant (Referee, credited) for the Romanian translation of the book (by Marius Cosmeanu).

REFERENCES

- Prof. Dr. Dragoș ȘTEFAN – Faculty of Mathematics and Computer Science,
  +40 755 119 883, dragos.stefan@fmi.unibuc.ro;
- Prof. Dr. Sorin DĂSCĂLESCU – Faculty of Mathematics and Computer Science,
  +40 727 072 968, sdascal@fmi.unibuc.ro;
- Lect. Dr. Roxana ZUS – Faculty of Physics,
  +40 722 586 420, roxanazus@gmail.com.